Data Analysis Report – Ames Housing Dataset

Introduction

In this project, I worked with the Ames Housing dataset, which contains information about house features and their sale prices. The main goal of this analysis was to understand the factors that affect house prices and prepare the data for further analysis and modeling

Data Cleaning Summary

During Phase 1, several data cleaning steps were performed to improve data quality. First, data types were corrected to ensure each column had the appropriate format. Missing values were handled using suitable strategies such as filling with median or mode depending on the feature. Duplicate rows were removed to avoid bias in the analysis. Additionally, outliers were detected and treated to reduce their impact on the dataset

Feature Engineering Summary

In Phase 2, multiple feature engineering techniques were applied. Categorical variables were encoded to make them usable for analysis. New domain-based features were created to better represent important aspects of the data. Interaction features were also added to capture relationships between variables. Log transformation was applied to reduce skewness in numerical features, especially the target variable. Some features were also grouped into bins, and highly correlated features were .removed to avoid redundancy

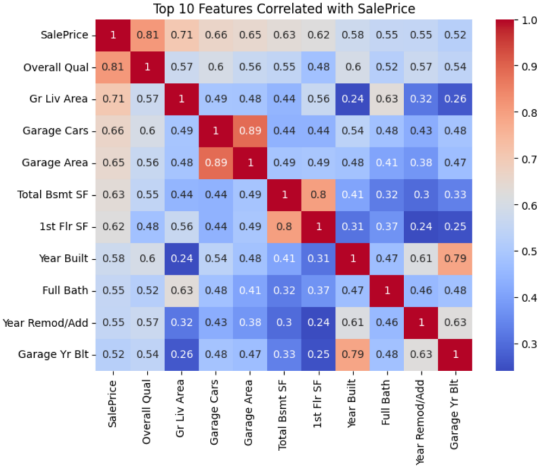
Key Findings

From the exploratory data analysis, several important insights were discovered

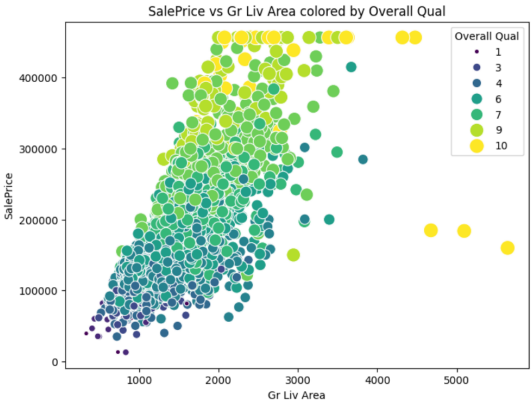
There is a strong positive relationship between overall quality and house price, meaning higher quality houses tend to have higher prices

Certain categories, such as neighborhoods or house conditions, show clear differences in average sale prices

The target variable (SalePrice) was initially skewed, .1 but after transformation, it became more normall .distributed, which improves analysis and modeling



The correlation heatmap shows that OverallQual has a strong positive correlation with SalePrice, making it one of the most important features



The scatter plot shows a clear positive relationship between house size and SalePrice, where larger houses tend to have higher prices

What I Would Do Next

If I had more time, I would build and evaluate machine learning models to predict house prices. I would also experiment with different algorithms and perform hyperparameter tuning to improve model performance